

Temperature Dependent Nickel Thin Film Chip Resistor (RTD)



FEATURES

- Alumina substrate base with nickel based PTC thin film element
- 0603, 0805, and 1206 sizes available
- Available in tape and reel packaging
- Standard R_{25} tolerances: $\pm 0.5\%$, $\pm 1\%$, $\pm 5\%$
- Operating temperature range: -55°C to $+150^{\circ}\text{C}$
- High stability over the entire temperature range
- C-UL-US recognized, file E148885
- AEC-Q200 qualified (grade 1)
- Material categorization: for definitions of compliance please see www.vishay.com/doc299912



LINKS TO ADDITIONAL RESOURCES



APPLICATIONS

Temperature compensation and sensing in

- Automotive
- Motor drives
- Lighting LED drivers
- Test and measuring equipment

QUICK REFERENCE DATA				
PARAMETER	VALUE			UNIT
DESCRIPTION	TFPT0603	TFPT0805	TFPT1206	
Resistance value at 25 °C ⁽¹⁾	100 to 1K	100 to 5K	100 to 10K	Ω
Tolerance on R_{25} -value	± 0.5; ± 1; ± 5			%
TCR at 25 °C	4110			ppm/K
Tolerance on TCR at 25 °C ⁽²⁾	± 400			
Operating temperature range: at rated power at derated power ⁽³⁾	-55 to +70 -55 to +150			°C
Storage temperature range	-55 to +150			
Dissipation factor δ (for information only) ⁽⁴⁾	1.8	2.3	4	mW/K
Maximum rated power at 70 °C (P_{70}) ⁽³⁾⁽⁴⁾	75	100	125	mW
Maximum working voltage RCWV ⁽⁵⁾	30	40	50	V
Weight	2	5.5	10	mg
Failure rate FIT _{observed}	≤ 0.1 x 10 ⁻⁹ /h			

Notes

- ⁽¹⁾ Other R_{25} -values are available upon request
- ⁽²⁾ Contact Vishay if closer TCR lot tolerance is desired
- ⁽³⁾ Derated power curve can be found in section "Power Derating". Power applied at maximum temperature should not let increase the film temperature by more than 1 K (1°C)
- ⁽⁴⁾ Valid for sensor element only in low dissipative mode. For dissipative mounting, please refer to APPLICATION INFORMATION
- ⁽⁵⁾ Rated continuous working voltage is maximum working voltage or $\sqrt{P_{70} \times R}$ whichever is less

APPLICATION INFORMATION

When the TFPT dissipates power, a temperature rise above the ambient temperature occurs, dependent on the thermal resistance of the assembled thermistor together with the mounting substrate. The (de)-rated power dissipation applies only if the long term permitted film temperature of 150°C is not exceeded by more than 1°C . Typically the thermal resistance (R_{thFA}) of a FR4 mounted TFPT0603 is around 250 K/W .

Please consider the application note "Thermal Management in Surface-Mounted Resistor Applications" (www.vishay.com/doc28844) for information on the general nature of thermal resistance.

STANDARD RESISTANCE VALUES at 25 °C in Ω

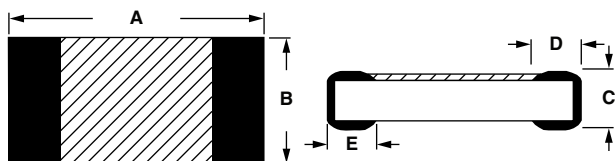
100	180	330	560	1.0K	1.8K	3.3K	5.0K	8.2K
120	220	390	680	1.2K	2.2K	3.9K	5.6K	10.0K
150	270	470	820	1.5K	2.7K	4.7K	6.8K	

GLOBAL PART NUMBER INFORMATION

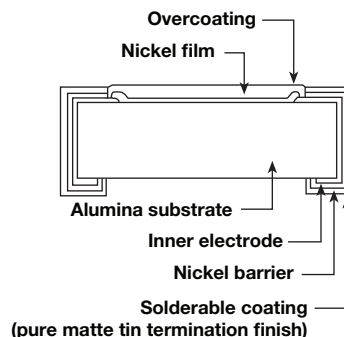
Global Part Numbering: TFPT1206L1002FM (preferred part number format)

T	F	P	T	1	2	0	6	L	1	0	0	2	F	M
GLOBAL MODEL				CHARACTERISTIC		RESISTANCE VALUE		TOLERANCE CODE		PACKAGING ⁽¹⁾				
TFPT0603 TFPT0805 TFPT1206				L = linear		1002 = 10K 1001 = 1K 1000 = 100R		D = ± 0.5 % F = ± 1 % J = ± 5 %		M = paper tape on reel, code ET1 (5000 pcs) V = paper tape on reel, code E52 (1000 pcs)				

Note
⁽¹⁾ According IEC 60286-3: 8 mm paper tape on $\varnothing$ 180 mm / 7" reel

DIMENSIONS in millimeters


PART NUMBER	A	B	C	D	E
TFPT 0603	1.55 ± 0.10	0.80 ± 0.10	0.45 ± 0.10	0.30 ± 0.20	0.30 ± 0.20
TFPT 0805	2.00 ± 0.15	1.25 ± 0.15	0.45 ± 0.10	0.40 ± 0.20	0.40 ± 0.20
TFPT 1206	3.05 ± 0.15	1.50 ± 0.15	0.55 ± 0.10	0.50 ± 0.25	0.50 ± 0.25

CONSTRUCTION

TESTS AND REQUIREMENTS

TEST	CONDITIONS ⁽¹⁾	REQUIREMENTS MAX. $ \Delta R_{25}/R_{25} $
High temperature exposure (storage)	AEC-Q200, 1000 h at 150 °C	0.25 %
Temperature cycling	AEC-Q200, 1000 cycles -55 °C / +125 °C	0.25 %
Biased humidity	1000 h, 1 mA biased at 85 °C / 85 % RH	0.25 %
	1000 h, 1 mA biased at 40 °C / 95 % RH	0.25 %
Operational life	1000 h, 10 % of P_{70} max biased at 85 °C	0.25 %
Mechanical shock	MIL-STD 202, method 213	0.25 %
Mechanical vibration	MIL-STD 202, method 204	0.25 %
Resistance to soldering heat	MIL-STD 202, method 210, condition K (reflow soldering)	0.25 %
ESD ⁽²⁾	AEC-Q200-002, HBM (CD) 0.2 kV (0603), 0.5 kV (0805), 0.5 kV (1206)	0.25 %
Board flex	AEC-Q200-005, 2 mm during 60 s	0.25 %
Terminal strength	AEC-Q200-006, shear test 17.7 N (0805, 1206) and 10 N (0603) during 60 s	0.25 %

Notes
⁽¹⁾ Environmental performance specifications use test procedures as outlined in MIL-R23648D, MIL-STD 202 and AEC-Q200

⁽²⁾ TFPTs are ESD sensitive

**AGENCY APPROVALS**

- C-UL certificate
- UL-US certificate

Note

- Agency approval documents, please see: www.vishay.com/ppg?33017&documents

AVERAGE RATIO R/R_{25} TFPT ALL SIZES AND VALUES									
TEMP.	R/R_{25}	TEMP.	R/R_{25}	TEMP.	R/R_{25}	TEMP.	R/R_{25}	TEMP.	R/R_{25}
		-20	0.825	20	0.980	60	1.150	100	1.337
		-19	0.828	21	0.984	61	1.155	101	1.342
		-18	0.832	22	0.988	62	1.159	102	1.347
		-17	0.836	23	0.992	63	1.164	103	1.352
		-16	0.839	24	0.996	64	1.168	104	1.357
-55	0.702	-15	0.843	25	1.000	65	1.173	105	1.362
-54	0.705	-14	0.847	26	1.004	66	1.177	106	1.367
-53	0.708	-13	0.851	27	1.008	67	1.182	107	1.372
-52	0.712	-12	0.854	28	1.012	68	1.186	108	1.377
-51	0.715	-11	0.858	29	1.017	69	1.191	109	1.382
-50	0.719	-10	0.862	30	1.021	70	1.196	110	1.387
-49	0.722	-9	0.866	31	1.025	71	1.200	111	1.392
-48	0.725	-8	0.869	32	1.029	72	1.205	112	1.397
-47	0.729	-7	0.873	33	1.033	73	1.209	113	1.402
-46	0.732	-6	0.877	34	1.037	74	1.214	114	1.407
-45	0.736	-5	0.881	35	1.042	75	1.219	115	1.412
-44	0.739	-4	0.885	36	1.046	76	1.223	116	1.417
-43	0.743	-3	0.889	37	1.050	77	1.228	117	1.422
-42	0.746	-2	0.892	38	1.054	78	1.232	118	1.427
-41	0.749	-1	0.896	39	1.059	79	1.237	119	1.432
-40	0.753	0	0.900	40	1.063	80	1.242	120	1.437
-39	0.756	1	0.904	41	1.067	81	1.246	121	1.442
-38	0.760	2	0.908	42	1.071	82	1.251	122	1.448
-37	0.763	3	0.912	43	1.076	83	1.256	123	1.453
-36	0.767	4	0.916	44	1.080	84	1.261	124	1.458
-35	0.771	5	0.920	45	1.084	85	1.265	125	1.463
-34	0.774	6	0.924	46	1.089	86	1.270	126	1.468
-33	0.778	7	0.927	47	1.093	87	1.275	127	1.473
-32	0.781	8	0.931	48	1.097	88	1.280	128	1.478
-31	0.785	9	0.935	49	1.102	89	1.284	129	1.484
-30	0.788	10	0.939	50	1.106	90	1.289	130	1.489
-29	0.792	11	0.943	51	1.110	91	1.294	131	1.494
-28	0.796	12	0.947	52	1.115	92	1.299	132	1.499
-27	0.799	13	0.951	53	1.119	93	1.303	133	1.505
-26	0.803	14	0.955	54	1.124	94	1.308	134	1.510
-25	0.806	15	0.959	55	1.128	95	1.313	135	1.515
-24	0.810	16	0.963	56	1.133	96	1.318	136	1.520
-23	0.814	17	0.967	57	1.137	97	1.323	137	1.526
-22	0.817	18	0.971	58	1.141	98	1.328	138	1.531
-21	0.821	19	0.975	59	1.146	99	1.333	139	1.536

RATIO FORMULA

$$R_T = R_{25} \times (9.0014 \times 10^{-1} + 3.87235 \times 10^{-3} (^\circ\text{C})^{-1} \times T + 4.86825 \times 10^{-6} (^\circ\text{C})^{-2} \times T^2 + 1.37559 \times 10^{-9} (^\circ\text{C})^{-3} \times T^3)$$

$$T(^\circ\text{C}) = 28.54 \times (R_T/R_{25})^3 - 158.5 \times (R_T/R_{25})^2 + 474.8 \times (R_T/R_{25}) - 319.85$$

RATIO TOLERANCES		
LOW TEMP.	HIGH TEMP.	TOL.
-55 °C	+150 °C	± 4 %
-40 °C	+125 °C	± 3 %
-20 °C	+85 °C	± 2 %
0 °C	+55 °C	± 1 %
+12 °C	+40 °C	± 0.5 %

RATIO TOLERANCE EXAMPLES:

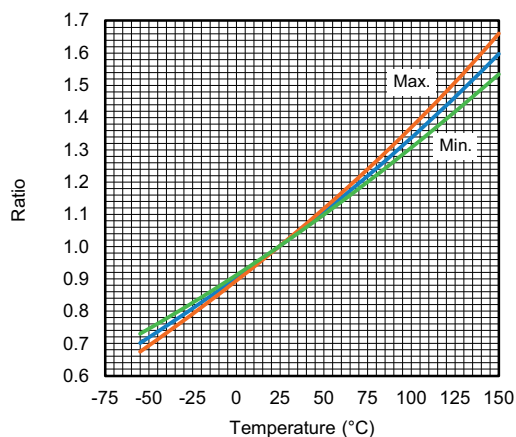
At 40 °C, ratio = 1.063 ± 0.5 % (0.005)
so, ratio = 1.058 to 1.068

At 125 °C, ratio = 1.460 ± 3 % (0.044)
so, ratio = 1.416 to 1.504

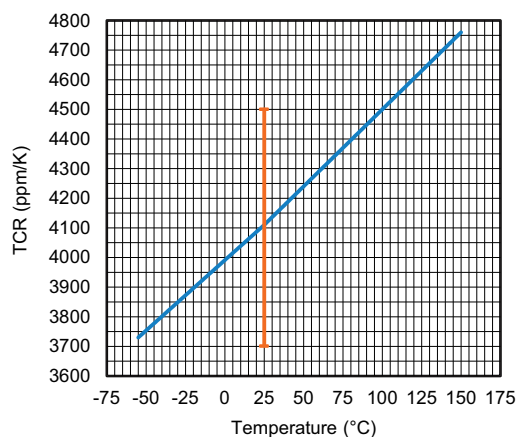
At intermediate temperatures, the ratios can be gradually adapted, for example at 105 °C the ratio tolerance will be ± 2.5 %.

For total resistance tolerance, the specific R_{25} tolerance needs to be multiplied with the ratio tolerance, for example a 100R 1 % at 25 °C will have a maximum resistance at 125 °C of 100R x 1.463 x 1.03 x 1.01 = 152.2 Ω.

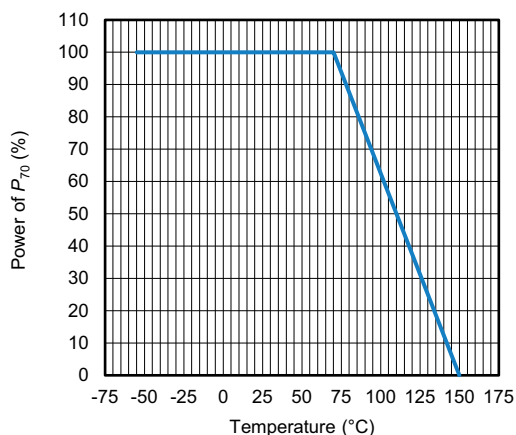
RATIO R_T/R_{25}



TCR TYPICAL VALUE



POWER DERATING



Note

- Zero (0 %) power is considered as measuring power that will generate a maximum film temperature increase of 1 °C



Disclaimer

ALL PRODUCT, PRODUCT SPECIFICATIONS AND DATA ARE SUBJECT TO CHANGE WITHOUT NOTICE TO IMPROVE RELIABILITY, FUNCTION OR DESIGN OR OTHERWISE.

Vishay Intertechnology, Inc., its affiliates, agents, and employees, and all persons acting on its or their behalf (collectively, "Vishay"), disclaim any and all liability for any errors, inaccuracies or incompleteness contained in any datasheet or in any other disclosure relating to any product.

Vishay makes no warranty, representation or guarantee regarding the suitability of the products for any particular purpose or the continuing production of any product. To the maximum extent permitted by applicable law, Vishay disclaims (i) any and all liability arising out of the application or use of any product, (ii) any and all liability, including without limitation special, consequential or incidental damages, and (iii) any and all implied warranties, including warranties of fitness for particular purpose, non-infringement and merchantability.

Statements regarding the suitability of products for certain types of applications are based on Vishay's knowledge of typical requirements that are often placed on Vishay products in generic applications. Such statements are not binding statements about the suitability of products for a particular application. It is the customer's responsibility to validate that a particular product with the properties described in the product specification is suitable for use in a particular application. Parameters provided in datasheets and / or specifications may vary in different applications and performance may vary over time. All operating parameters, including typical parameters, must be validated for each customer application by the customer's technical experts. Product specifications do not expand or otherwise modify Vishay's terms and conditions of purchase, including but not limited to the warranty expressed therein.

Hyperlinks included in this datasheet may direct users to third-party websites. These links are provided as a convenience and for informational purposes only. Inclusion of these hyperlinks does not constitute an endorsement or an approval by Vishay of any of the products, services or opinions of the corporation, organization or individual associated with the third-party website. Vishay disclaims any and all liability and bears no responsibility for the accuracy, legality or content of the third-party website or for that of subsequent links.

Vishay products are not designed for use in life-saving or life-sustaining applications or any application in which the failure of the Vishay product could result in personal injury or death unless specifically qualified in writing by Vishay. Customers using or selling Vishay products not expressly indicated for use in such applications do so at their own risk. Please contact authorized Vishay personnel to obtain written terms and conditions regarding products designed for such applications.

No license, express or implied, by estoppel or otherwise, to any intellectual property rights is granted by this document or by any conduct of Vishay. Product names and markings noted herein may be trademarks of their respective owners.